TenderLens

Single-Bid Risk in Assam Government Tenders

Predict • Explain • Check first

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Agenda

1. **The problem**
   Tenders and the one-bid risk
2. **The data**
   Source and what was usable
3. **Preparing data**
   Audit, cleaning, EDA, clues
4. **Modelling**
   Algorithms, comparison, results
5. **The app**
   TenderLens and live demo
6. **Wrap-up**
   Challenges, limits, next steps

Public Procurement: How a Tender Works

A department needs work done

Roads, buildings, supplies

It posts a tender online

Value, dates, rules

Companies send their bids

Each quotes a price

The best fair price wins

Competition keeps it fair

Example: PWD needs a road repaired → posts it on the Assam e-tenders portal → 5 companies bid → the work goes to the best fair offer.

Single-Bid Tenders: A Competition Risk

Many bids

- Prices can be compared
- Companies cut prices to win
- Government gets a fair deal

Only one bid

- Nothing to compare the price with
- No reason to give a discount
- No choice for the government

In Assam, 2,044 of 6,790 tenders (30.1%) got only one bid. A warning sign, not proof of wrongdoing.

The Financial Impact

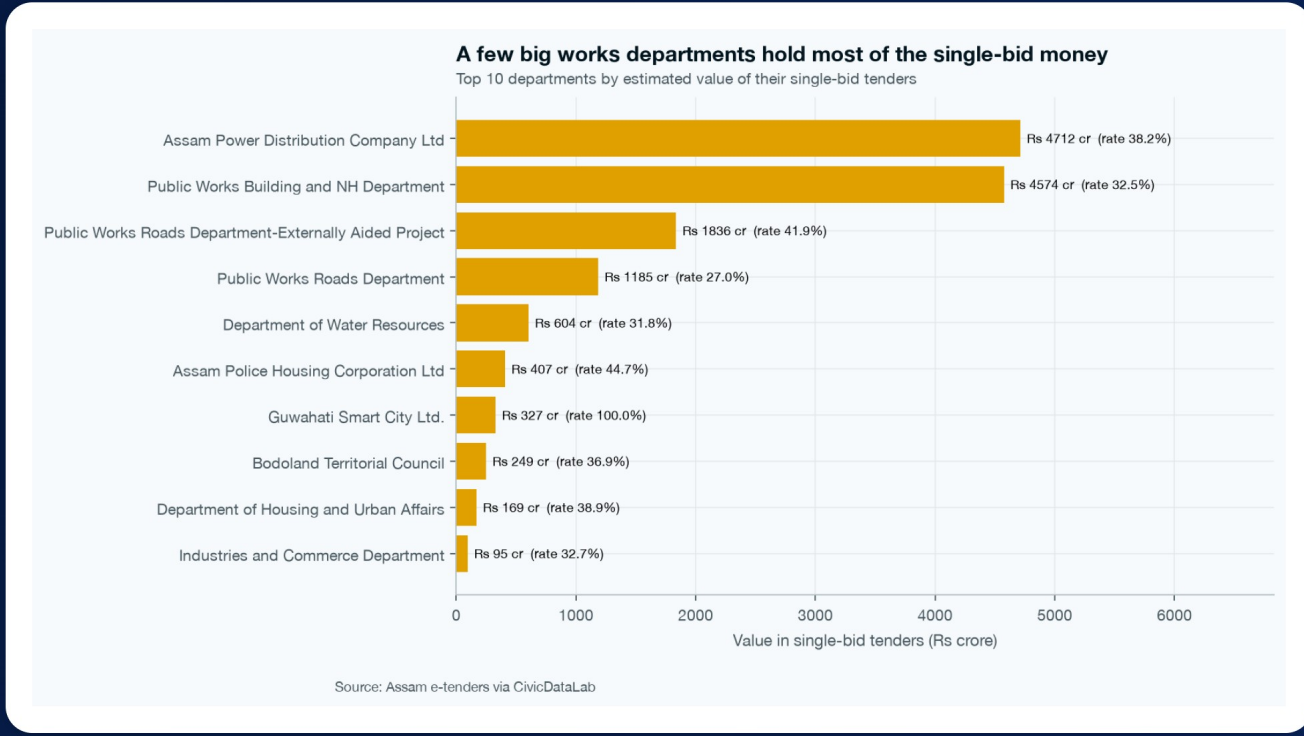
Rs 14,752 crore

went out with only one bidder

31%
of the tenders

36.5%
of the money

- One-bid works are the bigger ones
- 3 departments hold 75% of it
- That is where checks matter most



5,771 labelled tenders with a written value, total Rs 40,450 crore

Competition and Price: Discounts Begin at Four Bidders



96%

of the estimate paid with 1 bid

~100%

with 2 or 3 bids

90%

with 4+ bids: about 10% saved

Goal: real competition, not just 2 bids

Median awarded price ÷ estimate, 2,933 awarded tenders · a sign, not proof

Problem statement



Which newly published tenders are likely to get only one bid, why, and which should an auditor check first?

A single bid is seen only **after** bids open: too late to act

Auditors can't check every tender: Dec 2022 alone had 1,200+

They need a ranked list, with reasons, on publishing day

Project Objectives

Predict

Score each new tender’s single-bid risk on the day it is published

Explain

Show the reasons behind every score in plain words

Flag

Point out tenders that look unusual compared with normal ones

Build a tool

An app with a check-first list and an AI-written review brief

Screening support for auditors, not fraud detection

Data Source: Assam e-Tenders Portal

Source

CivicDataLab (open data on GitHub)

Period

April 2016 – May 2023

Size

39,500 CSV files · 72 MB

Format

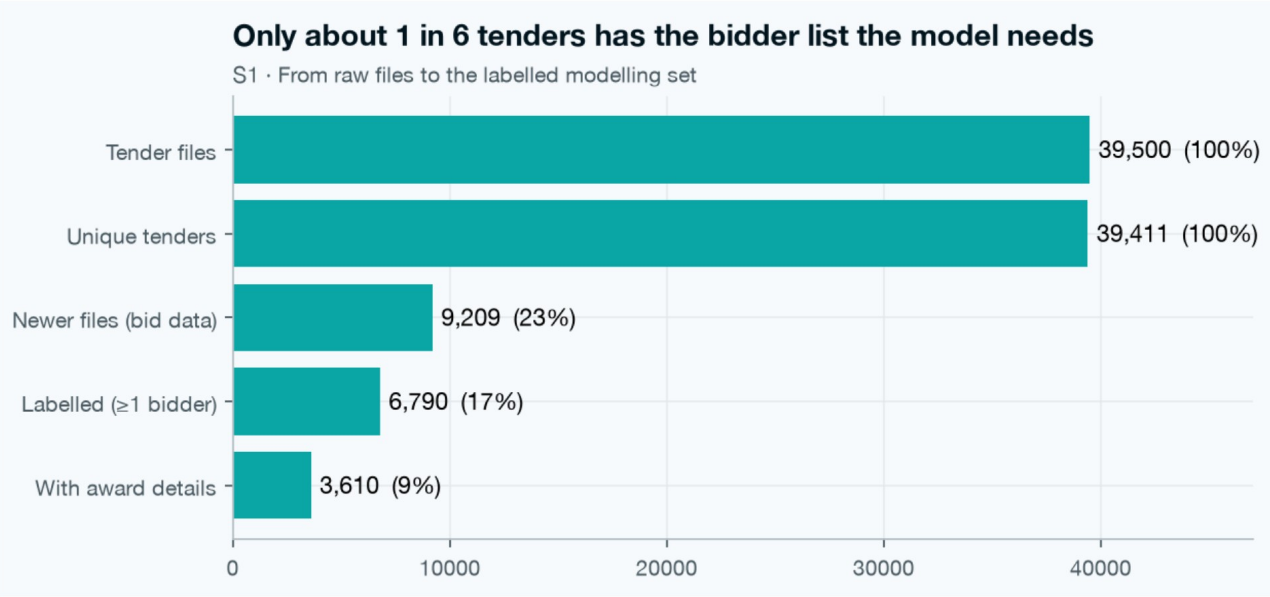
One file = one tender

What each file holds

- Department, title, category
- Tender value, EMD, fee
- Published, closing, opening dates
- Bidders list (newer files)
- Winner and awarded price
- Cancellations and amendments

The official OCDS feed was also checked, but it has no bid counts, so it was not used.

Data Funnel and Target Definition



- 39,411 unique tenders after removing 89 duplicates
- **6,790** have a verified bidder list: my modelling set
- Target: 1 = exactly one bid, 0 = two or more
- Unknown bids stay unknown, never counted as 0

☆ **2,044 single-bid tenders = 30.1% of the labelled set**

Tools and Technologies

Data work
Python 3.11,
pandas, NumPy

SQL
SQLite database for
EDA queries

Charts
Matplotlib, Seaborn:
85 charts

Machine learning
scikit-learn, XGBoost

Text (NLP)
TF-IDF, Sentence
Transformers (MiniLM)

Explain
SHAP, permutation
importance

Deep learning
TensorFlow 2.16 with
Keras 3

GenAI
LLM API (Gemini / Groq)
+ template

App
Streamlit, Plotly, fpdf2

Workspace
VS Code, Jupyter on
MacBook

Sharing
GitHub

Each tool used for the job it does best

Methodology: A 10-Step Pipeline

1
Data audit
Understand the files

2
Cleaning
One clean table

3
EDA + SQL
Find the patterns

4
Features
7 clues + time split

5
Models
Compare vs benchmarks

6
Text (NLP)
Do the words help?

7
Final test
One fair test + SHAP

8
Anomalies
Odd tenders, money

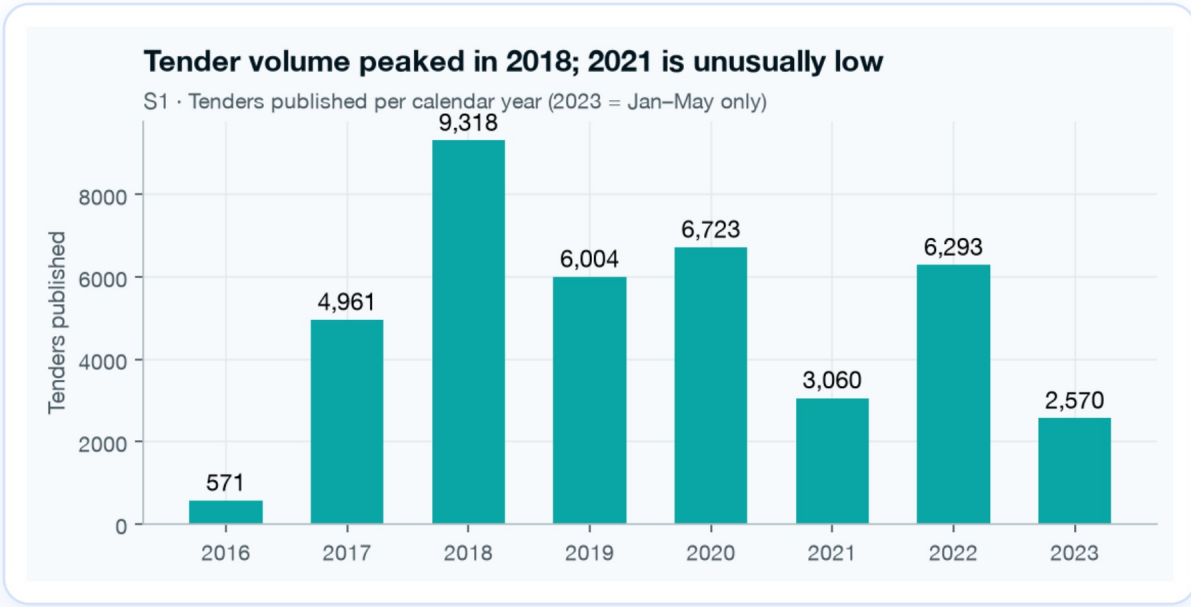
9
Deep learning
Does an ANN help?

10
App + GenAI
TenderLens tool

STEP 1 OF 10

Data Audit and File Structure

- First row = tender details; extra rows = lists of bidders and documents
- 30,291 older files have no bidder list
- 9,209 newer “final_” files hold the bids
- One column = winner only, another = full bidder list, so I counted bids from the full list



STEP 2 OF 10

Data Cleaning and Preparation

39,500 → 39,411
89 duplicate tender IDs removed

454 → 105
Department names merged

Rs 0 → blank
Plus a “value not written” flag

Text → dates
Numbers and 3 date columns fixed

19 + 129
Negative bid windows blanked, very long ones marked

Data gap
Mar 2021 – Feb 2022 marked, not a real drop

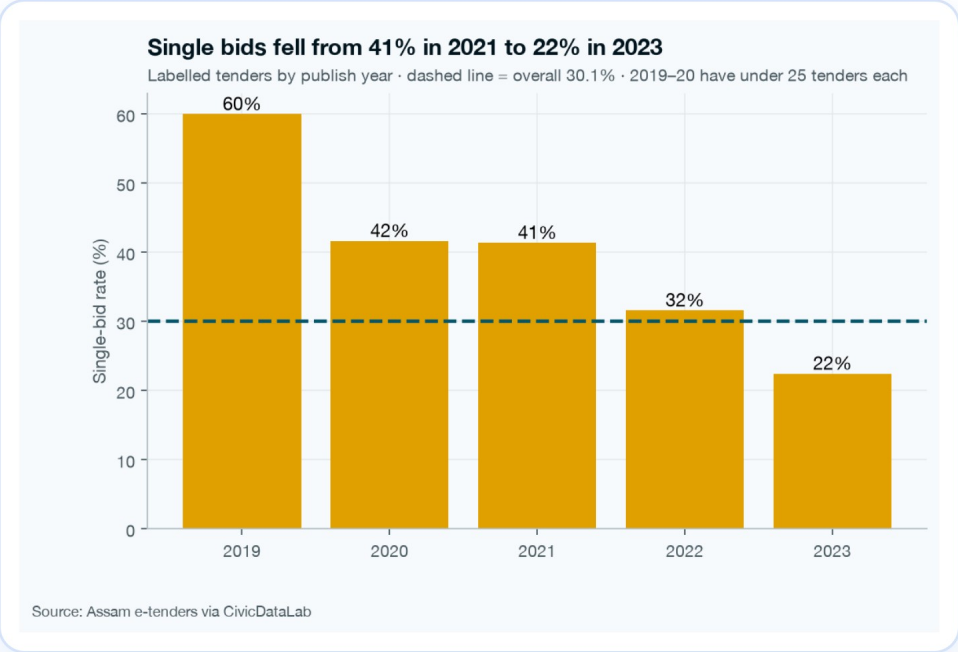
6,092
Re-tenders marked

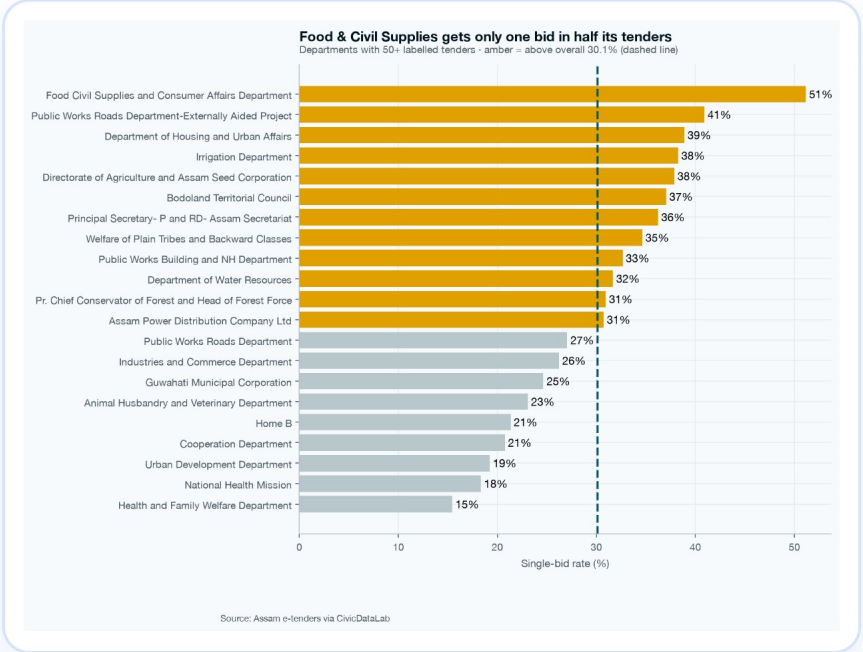
Label made
2,044 single-bid · 4,746 multi-bid

Output: one clean table · 39,411 tenders × 39 columns

STEP 3 OF 10 · PYTHON + SQL (SQLITE)

Exploratory Data Analysis (EDA)





Falling: 41% (2021) → 22% (2023)

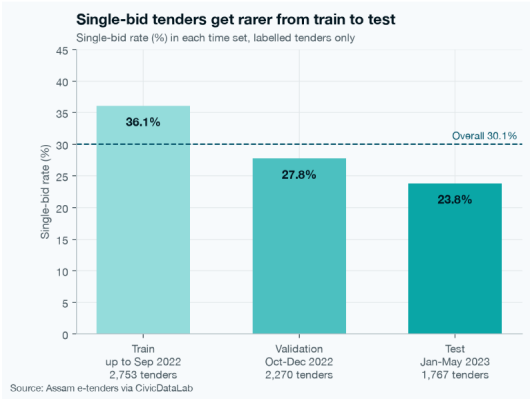
Departments: 51% vs 15%

Quick re-tender: 40%, the highest

STEP 4 OF 10

Feature Engineering and Time-Based Split

Clue	What it means
Department rate	Department's past single-bid rate
Tender value	Value on a log scale
Value not written	1 if the value is blank
Days to bid	Days firms get to send bids
Months to year-end	Months left until 31 March
Same-day lot	Same title posted the same day
Quick re-tender	Same work re-posted within 60 days



Train to Sep 2022 · **Validation** Oct–Dec 2022 · **Test** Jan–May 2023

Only facts known on publishing day

Department rate from earlier tenders only

Blanks filled with training medians

Algorithms Used

Logistic Regression Main
Weighs the 7 clues into one risk score

Random Forest
Many decision trees vote together

XGBoost
Trees that learn from earlier trees' mistakes

TF-IDF
Counts the important words in a title

MiniLM embeddings
Turns a title into 384 numbers of meaning

SHAP
Shows how much each clue pushed a score

Isolation Forest
Finds tenders that are easy to separate: unusual

Neural network (ANN)
Layers of neurons learn deeper patterns

Autoencoder
Rebuilds normal tenders; a bad rebuild means odd

Predict: LogReg, RF, XGBoost, ANN

Text: TF-IDF, MiniLM

Explain / odd: SHAP, IF, autoencoder

Model Evaluation Framework

Two benchmarks to beat

Random checking, and Rule B: quick re-tenders first, then department history

The right scores

PR-AUC, ROC-AUC, and how many single bids are caught in the riskiest 20%

A fair setup

Time split, class weights for imbalance, plan locked before the test, test used once

Proof it isn't luck

500 re-draws (bootstrap), several seeds, results matched on two machines

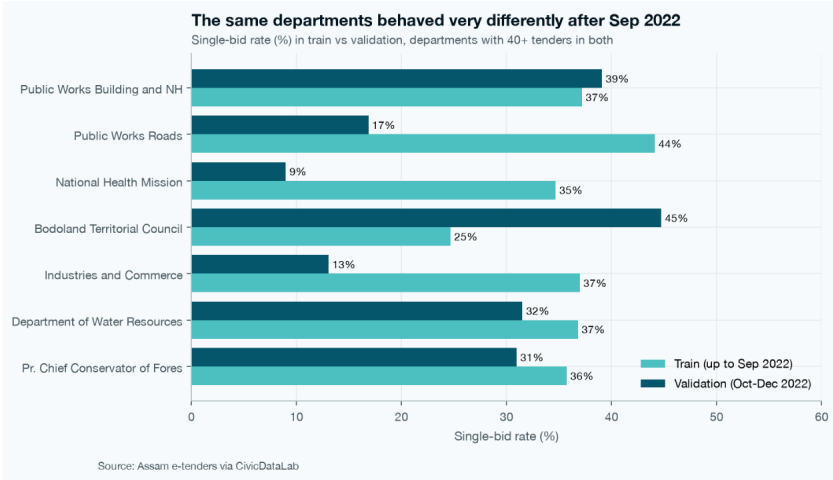
Why not accuracy? Saying "no" to every tender already scores 76%

STEP 5 OF 10

Baseline Models and Data Drift

Validation (Oct–Dec 2022)	PR-AUC
Random checking	0.288
Rule B (simple rule)	0.268
Logistic Regression	0.252
Random Forest	0.233
XGBoost	0.223

Higher is better · more clues (34) did not help either



⚠ **Drift:** 21 of 34 clues flipped after Sep 2022

🔍 A model could tell old from new tenders (0.818), so the past ≠ the future

STEP 6 OF 10

Text Analytics (NLP) and Label Timing

Do the words help?

TF-IDF 0.274 · MiniLM embeddings 0.300 (best text method)

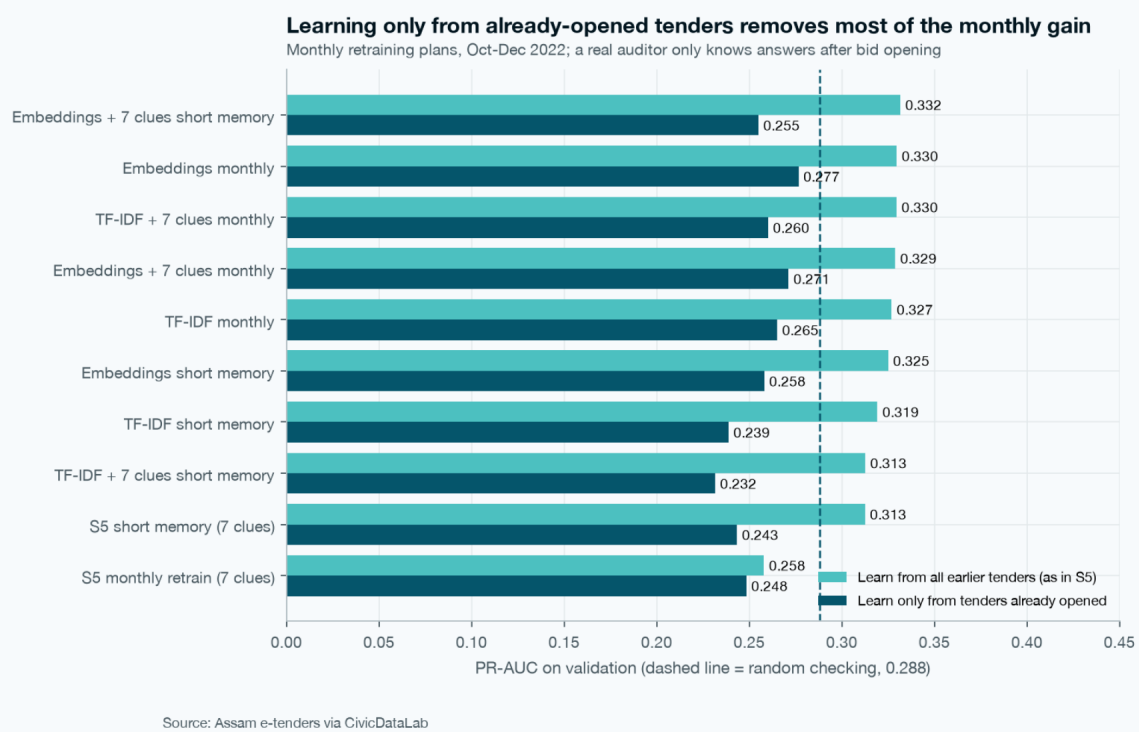
But it beat random in only 75% of re-draws, so not proven

Words drifted too: old vs new scheme names (0.913)

Label timing check

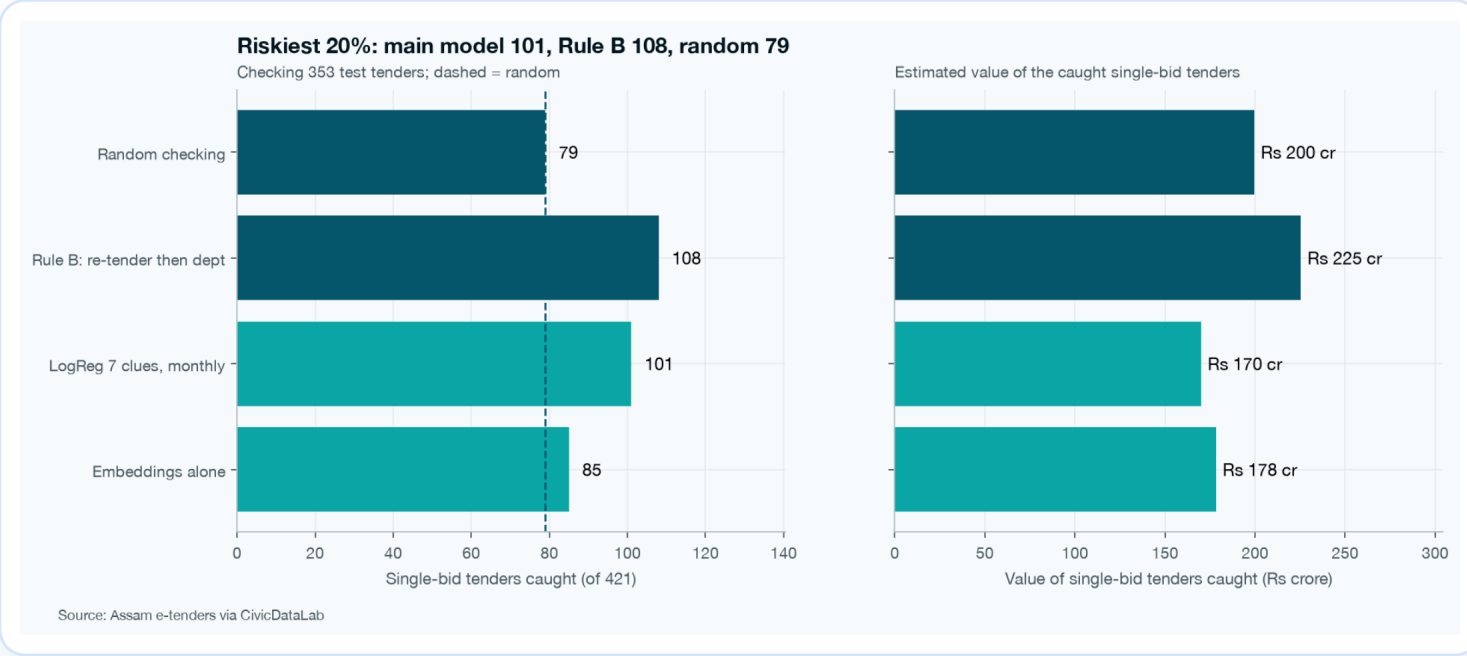
A model may learn only from tenders whose bids are already opened

1,098 of 2,270 opened later: fixing this removed a fake gain



STEP 7 OF 10 · JAN–MAY 2023 · 1,767 TENDERS

Final Test Results on Unseen Data



Plan locked, test used once

+22 found, same effort

Won **95.8%** of 500 re-draws

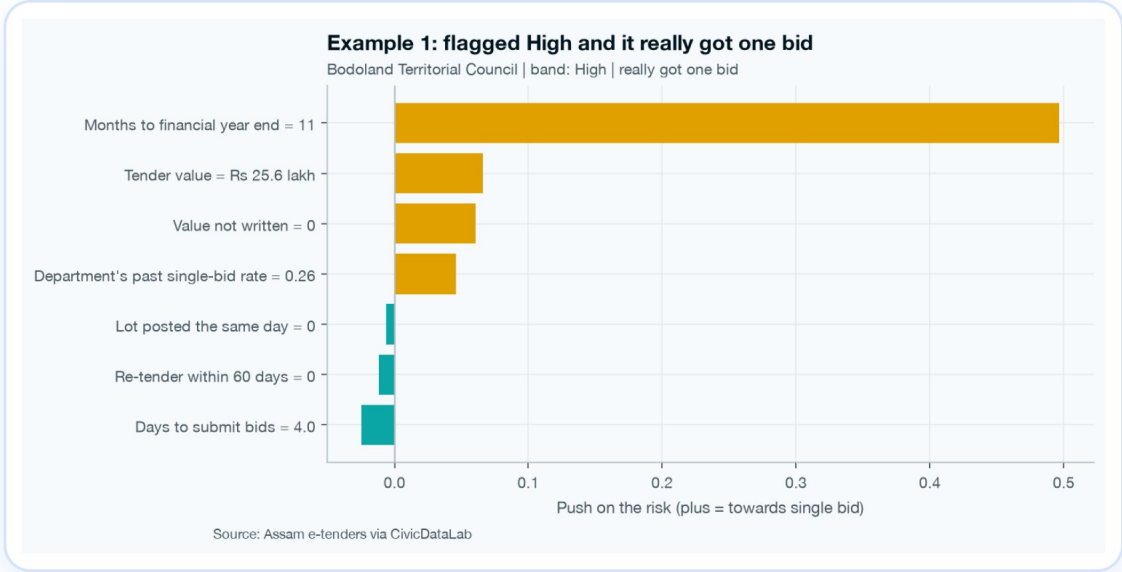
Rule B found 108: reported honestly

STEP 7 OF 10 · EXPLAINABILITY

Model Explainability with SHAP



All tenders: which clues matter most



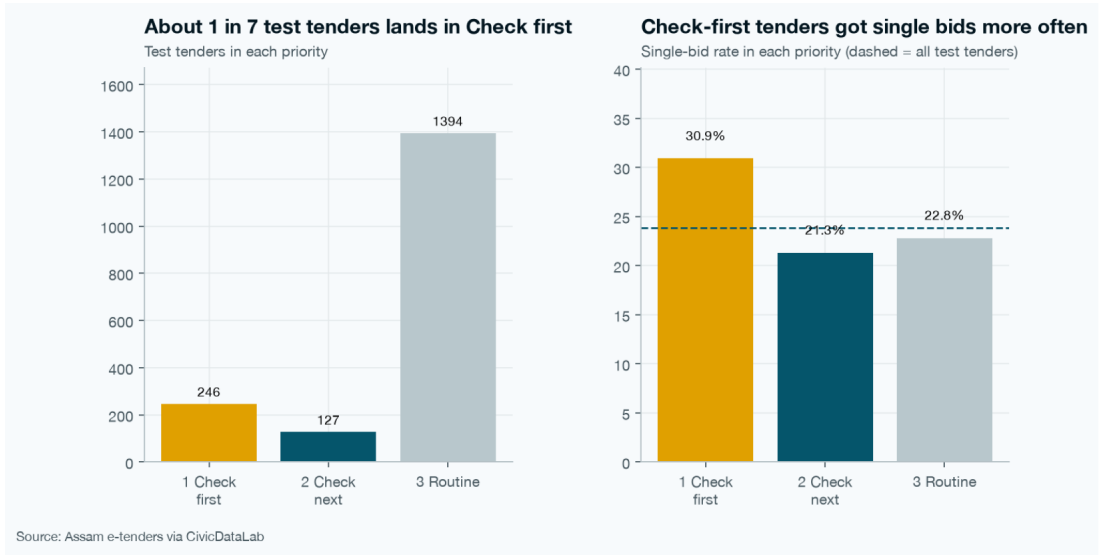
One tender: why it was flagged High

- Early in the year (Apr–May) → risk up
- Value not written → risk down
- Quick re-tender → risk up

STEP 8 OF 10

Anomaly Detection and Prioritisation

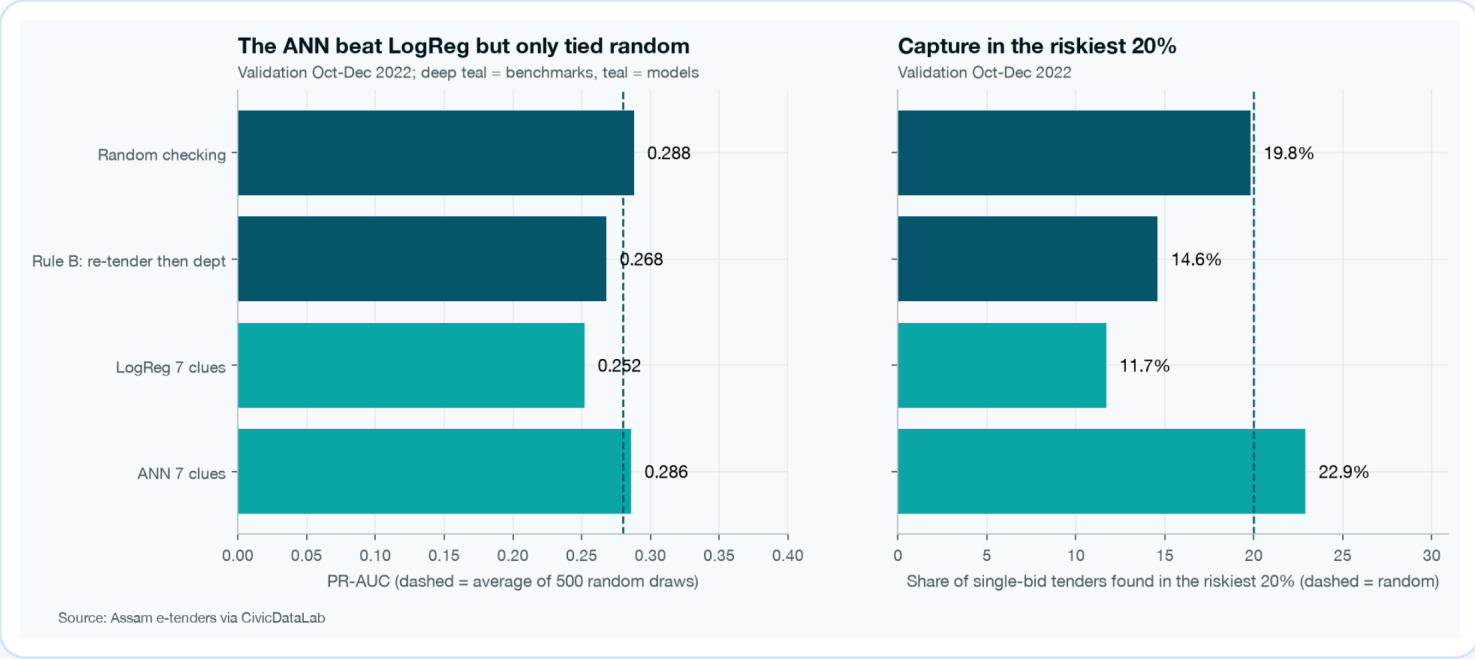
- Isolation Forest flags the oddest 5% and names the odd clue
- 54 of 74 unusual tenders were also High risk
- Caught a data error: a solar plant valued at Rs 3,000
- Repeat winners: 37.8% of single-bid awards vs 26.2%
- Re-posting didn't help: 19% → 33% single-bid



Check first = High risk + (unusual or ≥ Rs 1 crore): 246 tenders, about 1 in 7, 30.9% single-bid

STEP 9 OF 10 · TENSORFLOW / KERAS

Deep Learning Comparison



ANN 0.286 vs random 0.288

Beat random in only 43% of re-draws

Autoencoder over-flagged: 221 vs 74

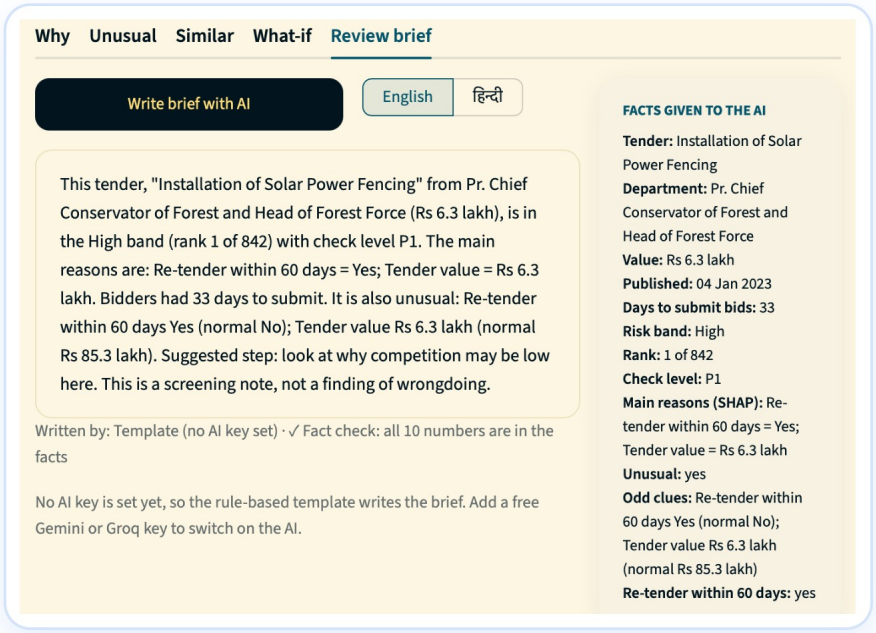
LogReg + Isolation Forest stay

STEP 10 OF 10 · STREAMLIT

Application and GenAI Review Brief



Overview: January 2023



Review brief: written only from verified facts

- 8 pages, ranked inside each month
- Bands High / Medium / Low, not %
- LLM brief, every number fact-checked
- Model health: monthly drift check

Live Demonstration

TenderLens in action

01 Overview: January 2023

02 Risk list and the audit budget

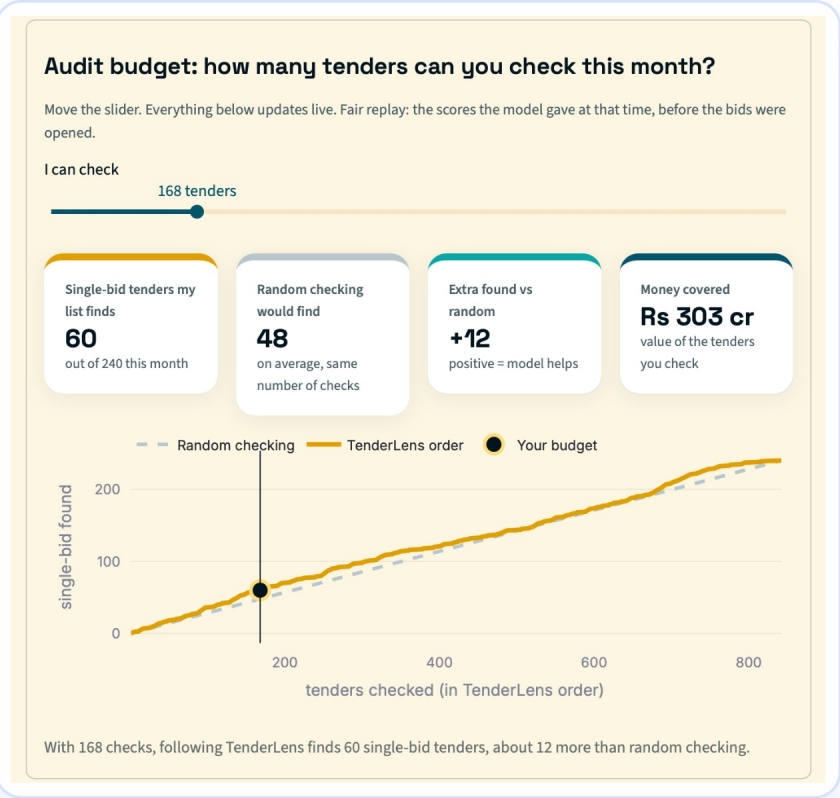
03 One tender: Why, Unusual, Brief

04 Department report card

05 Model health

06 Score a new tender

Business Impact and Use Cases



January 2023 · 168 checks

60 vs 48

found by TenderLens vs random · +12 · Rs 303 crore

Who can use it

- Audit and vigilance teams: a check-first list each month
- Department heads: act before bids close
- Finance offices and data quality teams

Challenges and Solutions

⚠ Messy file structure

Studied files by hand; counted bids from the full bidder list

⚠ Gap in the data

Marked Mar 2021 – Feb 2022 instead of reading it as a drop

⚠ Drift after Sep 2022

Time split, honest tests, monthly retraining plan

⚠ Hidden timing leak

Learn only from tenders whose bids were already opened

⚠ Library clashes

NumPy, SHAP, TensorFlow, pyarrow: pinned safe versions

⚠ ANN stopped too early

Balanced sample weights fixed early stopping

Limitations

- **Short history:** most labelled tenders are from 2022
- **Drift:** patterns change, so the model must be retrained monthly
- **Small gain:** 101 vs 79, and a simple rule found 108
- **Weak clues:** local market and advertising are not in the data
- **Screening only:** a high score never means wrongdoing

Recommendations

1. **Use a monthly check-first list**
   Start with High risk + unusual or ≥ Rs 1 crore

2. **Aim for 4+ bidders**
   Longer windows and wider ads on big, high-risk works

3. **Watch departments going the wrong way**
   Bodoland Territorial Council: 24.7% → 39.9%

4. **Retrain monthly, fix data**
   Make the value and district fields required

Future Scope

Deploy online

Streamlit Cloud with a live AI key

More states

Test if the same clues work elsewhere

Auditor feedback

Learn from what checks actually found

Auto drift alerts

Retrain automatically every month

Full tender text

The portal cuts descriptions at 250 letters

District view

Clean the location field into districts

Conclusion: Key Takeaways

01 One-bid tenders: 30.1% of tenders, 36.5% of the money

02 Patterns changed after Sep 2022, so I tested only on the future

03 Final model: 101 found vs 79 by random, a small but real win

04 A simple, explainable model beat complex ones on this data

05 TenderLens turns it into a check-first list with reasons

Thank you!

Questions?

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